LECTURE 1: INTRODUCTION

Introduction

Welcome to lecture 1. In this lecture we are going to discuss the history of measurement and evaluation. We will look at the different contexts that is the African, Chinese, European and American context. We are also going to look at the general definition of education. Let us begin by looking at the learning outcomes below.

LEARNING OUTCOMES

After this lecture, the learner should be able to:

- Discuss how testing developed in China, Europe and America and its relationship to African context.
- Explain the concept of education with special reference to United Nations, World Trade Organization and the Kenyan constitution

BRIEF HISTORY OF TESTING AND MEASUREMENT

The use of tests for identifying potential employees; professional trainees' selection, readiness and placement is a long-established practice in human history.

African Context

Did the pre-modern African society have tests? Yes. Wrestling, jumping, dancing, tooth extraction/removal, and circumcision, were among the tests that were used by women for the choice and the selection of suitors; show of bravery; and readiness for the society's defence.

In China, Europe and USA the increase use of testing was attributed to three major areas of development: civil-service; school; and study of individual differences**.**

Chinese Context

- Civil-service testing began in China about 3000 years ago when an emperor decided to assess the competency of his officials. It has been reported that by the year 2200 B.C, the emperor of China examined his officials every third year and after three examinations, he either promoted or dismissed them from Civil Service.
- Later, the Chinese government positions were filled by persons who scored well on examinations that covered topics such as music, horsemanship, civil law, and writing.
- Such examinations were eliminated in 1905 and were replaced by formal educational requirements.

European Context

- Students in European schools were given oral examinations until after the 12th century.
- In the 16th century the Jesuits (catholic order/organization that vow for poverty and obedience) started using tests for the evaluation and placement of their students. By this time, Jesuits were running schools across Europe. The Jesuits' standardized curriculum and teaching methods became the basis of many education systems to-day.
- In 19th Century the study of individual difference led by Sir Francis Galton (1822-1911) began in Great Britain. He was the first experimental psychologist to look into psychological differences (nature and nurture) in sensory and motor skills between people

and the application of statistical methods (correlation and regression) in the analysis and quantification of the individual differences.

- By 1905 when China was phasing out the use of examinations in public service, civil-service examinations were being developed in Britain and the United States as way of selecting applicants for government jobs.
- In 1905 Alfred Binet (1871-1938) developed the first individual tests of intelligence as part of his work on the study of individual differences (M.A/C.A x 100). He was asked by the French government to come up with a test of identifying children who were not benefitting from school instructions.

American Context

- During War 1(1914-1918) United States Army used tests for selection of army recruits.

https://www.youtube.com/watch?v=NValmBwli1Q

https://www.youtube.com/watch?v=FOmGF_gJ_6o

https://www.youtube.com/watch?v=fWb12WjiiO8

WHAT IS EDUCATION?

Though education is seen as both an investment and a social service to the community and the nation, it means different things to different people or organizations.

1. Generic (general) Definition

From a generic perspective, education simply means:

- Transmission of knowledge, skills and attitudes from one person to another or the transmission of knowledge from a teacher to a pupil.
- The transmission of one's culture from generation to generation or process by which people are prepared to live effectively in their environment. [Refer to recent Masaai Moran graduation]

Education can therefore be seen as:

- The process of developing the capacities and potentials of the individual so as to prepare that individual to be successful in a specific society or culture. Education in this context is a continuous process which begins at birth and continued throughout life.
- It is a process of bringing about a permanent change in human behavior.

From these perspectives, education has two dimensions:

(i) The content (what is taught).

(ii) The process / method of imparting knowledge or molding individuals in society in order to develop their potentials (how it is taught). This is the process of imparting knowledge e.g. through imitations, storytelling etc.

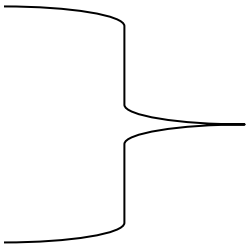
Education as a process involves teaching and learning which may occur in a <u>formal setting</u> as in a school, in <u>non-formal</u> settings as in artisan training centers, or in an <u>informal setting</u> as in families, bush schools and peer groups.

Ref: Hidden curriculum using iceberg example:

Iceberg

Formal education

Non-formal education



Informal education Hidden curriculum

LEARNING AND TESTING IN AFRICAN INDIGENOUS EDUCATION CONTEXT

1. **Learning Through Productive Work**

   Children learn the right type of masculine or feminine roles through doing and working hand in hand with adults.

   i. Boys with their fathers
   ii. Girls with their mothers

   Their roles were given according to the age of the learner–from simple to complex duties.

   The learning process involved:

   a) Seeing (Observation)
   b) Imitation

2. **Learning Through Oral Literature**

   Through writing was unknown during pre-colonial society learning took place through:

   i. Myths = tales of imaginary description of things.
   ii. Legends = tales of fabricated accounts.
   iii. folk-songs and dances = dances during rite of passage, funerals

3. **Learning Through Ceremonies**

   i. Rite of passage
   ii. During marriage, etc

4. **Learning Through Hunting**

   – Hunting: mockery and real hunting
   – Trapping animals

5. **Learning Through the Inculcation of Fear**

- Discouraging females not to eat eggs
- Denial of food for allowing bulls to fight.
- Being punished for misconduct and carelessness

6. **Learning Through Formal Teaching**
    - Though there was no structured curriculum, learning through apprenticeship was a formal and direct e. g. learner being sent by the parents to work with craftsmen, potters, blacksmiths, and stool makers. The same was true with the acquisition of hereditary occupations e.g. a herbalist in handing over his "secrets" about what medicine to use for which disease and how would instruct his learnerfrom time to time until he became knowledgeable and proficient in its practice.

So long as learning was taking place, testing of the same though not in writing or by use of paper and pen, was also in place. Punishment and corrections were indicators of failure of acquisition of expected knowledge, skills and values. Praises and selection for leadership positions were indicators of one having passed or acquired expected knowledge skills and values of the society.

[Source: African Indigenous Education: As practiced by the Acholi of Uganda by J.P. Ociti].

2. United Nations Definition

According to United Nations, education is ***a common good***. It is a common good in the sense that it is beneficial for all members of a given community. Education helps a community to escape poverty and create wealth. Education is a stepping stone through which students/learners can make something better for their future and their communities. It is a human right. This is why countries are encouraged to provide free basic education (primary and secondary education). Comment on World Bank funding primary instead higher education on the ground higher in primary education investment.

The late Nelson Rolihlanla Mandela described education as *"the most powerful weapon which you can use to change the world"***.**

The late Mwalimu Nyerere saw education as *a weapon for liberation and as a weapon to fight poverty* and not as a way to escape poverty.

3. *World Trade Organization (WTO)*

According to World Trade Organization (WTO), education is a saleable /tradable commodity. It is a commodity like hamburger, goats etc that are put on market for sale.

4. *Our Constitution*

Our constitution conceptualizes education as a fundamental human right. Section 43 (f) states that every person has the right to education. Section 53 states that every learner has the right to free and compulsory basic education.

Classification of Education

1. *By Levels*

Three levels of education in Kenya are:

- Primary
- Secondary

} UNESCO defines these two levels as Basic Education

- Tertiary
  - Middle level
  - University level

2. *By Methods*

- Formal – School settings
- Non-formal – By attachment
- Informal – Family settings

3. ***By Programs*** include:

- Special education
- Science and technology education
- Arts / humanities education
- Vocational education

Economy Rationale for Classifying Education

The classification of modern education by levels, programs and methods enables an investor (parents, ministry, etc) to set;

i. clear objectives
ii. to make clear assessment of the investment requirements in terms of:
    a. material and human resources, as well as
    b. the demand for any particular type or level of education. This is where assessment or measurement of objectives comes in.
    https://www.youtube.com/watch?v=FOmGF_gJ_6o
    https://www.youtube.com/watch?v=KUiTUbjljmU

review questions

a) discuss how testing developed in Europe, America and Asia

b) discuss how the following entities describe the concept of education

Further Reading

1. Education Measurement and Evaluation : J.Swarupa Rani, Discovery Publishing house.

2. Measurement and Evaluation in Teaching : Norman Edward Gronlund Macmillian

3. Measurement and Assessment in Teaching : Robert L. Linn Pearson Education India.

4. Program Evaluation and performance measurement : James C. Me. David, Laura R.L. Hawthorn, Sage Publications.